Online Supplementary Material

Order Cancellation-Aware On-Demand Ride-Pooling:
Formulation and Data-Driven Decomposition Algorithm

This document accompanies the main article. Sections S1–S4 provide the complete proof of Theorem 4, both NP-hardness proofs, worst-case loss methodology, predictor-sensitivity experiments, and algorithmic procedures. The full routing-pricing formulation is retained in Appendix G of the main article. Behavioral-model comparisons, simulation robustness checks, and the quadratic reformulation proof are retained in Appendices D–F of the main article. References to the main article retain its numbering; supplemental sections, equations, tables, and algorithms use the prefix S.

S1. Auxiliary Proofs

S1.1. Full Proof of Theorem 4

We first summarize the notation and separate structural discrepancy from prediction error. We then prove the bound in three steps and explain its numerical evaluation.

Notation used in the proof. For ease of reference, we summarize the matching decisions, profit measures, and error bounds below. A matching specifies which orders are assigned to each vehicle; fixing a matching still leaves routing and pricing to be optimized.

- $b \in \Omega$ denotes a feasible pooling pattern that assigns a set of orders $\mathcal{R}_b$ to one vehicle. The vector $\boldsymbol{\alpha} = (\alpha_b)_{b\in\Omega} \in \mathcal{A}$ represents a feasible matching, where $\alpha_b = 1$ if pattern b is selected and $\alpha_b = 0$ otherwise; $\mathcal{A}$ is the common feasible set of pattern selections.
- v_b is the optimal O-CAM profit for fixed pattern b, obtained by optimizing its route and prices. In contrast, η_b is its A-CAM profit, obtained by optimizing prices using the predicted tour length, average waiting-for-pickup time, and average detour ratio.
- η_b^0 is the A-CAM profit after replacing these predictions by the exact tour length ℓ_b^*, average waiting-for-pickup time $\bar{w}_b^*$, and average detour ratio $\bar{e}_b^*$ from a consistently chosen optimal fixed-pattern O-CAM solution, and reoptimizing prices. Because A-CAM still uses average service measures, η_b^0 need not equal v_b.
- $V(\boldsymbol{\alpha}) = \sum_b v_b \alpha_b$ is the optimal O-CAM profit for fixed matching $\boldsymbol{\alpha}$, whereas $\widehat{V}(\boldsymbol{\alpha}) = \sum_b \eta_b \alpha_b$ is the profit assigned to the same matching by A-CAM.
- $\boldsymbol{\alpha}^* \in \arg\max_{\boldsymbol{\alpha}\in\mathcal{A}} V(\boldsymbol{\alpha})$ is an optimal O-CAM matching (globally optimal matching), and $Z^* = V(\boldsymbol{\alpha}^*)$ is the corresponding optimal profit.
- $\widehat{\boldsymbol{\alpha}}$ is the feasible matching returned by the A-CAM solution procedure; it need not be globally optimal for A-CAM. The quantity $V(\widehat{\boldsymbol{\alpha}})$ is the profit obtainable by optimizing routing and pricing exactly while preserving this returned matching.
- $A_b = |v_b - \eta_b^0|$ measures the structural discrepancy remaining with exact tour length and average service measures. The bound $B_b^{\text{pred}} = C\epsilon_\ell + \sum_{r\in\mathcal{R}_b} p_r (L_w \epsilon_w + L_e \epsilon_e)$ controls the additional profit difference caused by prediction errors. Here, $(\epsilon_\ell, \epsilon_w, \epsilon_e)$ bound errors in tour length, average waiting-for-pickup time, and average detour ratio, respectively, and (L_w, L_e) are the corresponding sensitivity bounds for normalized order revenue with respect to pickup time and detour ratio.
- $E(\boldsymbol{\alpha}) = \sum_b (A_b + B_b^{\text{pred}})\alpha_b$ bounds the absolute difference between $V(\boldsymbol{\alpha})$ and $\widehat{V}(\boldsymbol{\alpha})$. Thus, $E(\boldsymbol{\alpha}^*)$ and $E(\widehat{\boldsymbol{\alpha}})$ evaluate the same error bound at the optimal O-CAM matching (globally optimal matching) $\boldsymbol{\alpha}^*$ and the matching $\widehat{\boldsymbol{\alpha}}$ returned by PAD^2 when solving A-CAM, respectively; neither is the actual optimality loss.
- $\Gamma \geq 0$ bounds the A-CAM optimization gap, $\max_{\boldsymbol{\alpha}\in\mathcal{A}} \widehat{V}(\boldsymbol{\alpha}) - \widehat{V}(\widehat{\boldsymbol{\alpha}})$. It accounts for incomplete optimization of A-CAM and can be set to zero when the returned matching is globally optimal for A-CAM.

Preliminaries: pattern profits and sources of error. We use the matching-indexed pooling patterns and notation of Section 4.5. For each $b \in \Omega$, the approximate profit is

$$\eta_b = \max_{\boldsymbol{u}\in[0,1]^{|\mathcal{R}_b|}} \left\{ \sum_{r\in\mathcal{R}_b} h_r(\widehat{w}_b, \widehat{e}_b, u_r) p_r u_r - C\widehat{\ell}_b \right\}, \tag{S1}$$

with pricing domain $\mathcal{U}_b = [0,1]^{|\mathcal{R}_b|}$. Replacing the predictions by $(\ell_b^*, \bar{w}_b^*, \bar{e}_b^*)$ gives the pricing problem defining η_b^0. Step 1 below establishes

$$|\eta_b - \eta_b^0| \le B_b^{\text{pred}} := C\epsilon_\ell + \sum_{r\in\mathcal{R}_b} p_r(L_w\epsilon_w + L_e\epsilon_e). \tag{S2}$$

The intermediate value η_b^0 separates the profit discrepancy into two components:

$$v_b - \eta_b = (v_b - \eta_b^0) + (\eta_b^0 - \eta_b).$$

The two components represent structural discrepancy and prediction error, respectively. Hence, the triangle inequality gives the pattern-level bound in Eq. (59):

$$|v_b - \eta_b| \le |v_b - \eta_b^0| + |\eta_b^0 - \eta_b| \le A_b + B_b^{\text{pred}}.$$

Illustration of matching-selection loss. The distinction between pattern-level profit error and the resulting matching loss is important: even after exact routing and pricing optimization, an inferior matching can remain suboptimal. For example, consider two orders and three feasible pooling patterns: b_1 assigns both orders to one vehicle, whereas b_2 and b_3 assign the two orders separately to two vehicles. Suppose their exact profits are $(v_{b_1}, v_{b_2}, v_{b_3}) = (20, 10, 8)$, while their approximate profits are $(\eta_{b_1}, \eta_{b_2}, \eta_{b_3}) = (17, 11, 8)$. A-CAM would select b_2 and b_3, whereas the globally optimal O-CAM solution selects b_1. The resulting optimality loss is $20 - (10 + 8) = 2$, which cannot be recovered by post-optimization because the matching remains fixed.

Proof of Theorem 4. With these preliminaries in place, we now prove Theorem 4 in three steps.

Step 1: Bound the profit error for each pattern. For a fixed pattern b, compare the two objectives at the same pricing vector $\boldsymbol{u} \in \mathcal{U}_b$:

$$f_b(\boldsymbol{u}) = \sum_{r\in\mathcal{R}_b} p_r u_r h_r(\widehat{w}_b, \widehat{e}_b, u_r) - C\widehat{\ell}_b,$$
$$f_b^0(\boldsymbol{u}) = \sum_{r\in\mathcal{R}_b} p_r u_r h_r(\bar{w}_b^*, \bar{e}_b^*, u_r) - C\ell_b^*.$$

The two expressions differ only in the values of average waiting-for-pickup time, average detour ratio, and P&D tour length; the prices are held fixed. Subtracting the two objectives and applying the triangle inequality, with $p_r \ge 0$ and $C \ge 0$, gives

$$\begin{aligned}
|f_b(\boldsymbol{u}) - f_b^0(\boldsymbol{u})| &= \left| \sum_{r\in\mathcal{R}_b} p_r \left[u_r h_r(\widehat{w}_b, \widehat{e}_b, u_r) - u_r h_r(\bar{w}_b^*, \bar{e}_b^*, u_r) \right] - C(\widehat{\ell}_b - \ell_b^*) \right| \\
&\le C|\widehat{\ell}_b - \ell_b^*| + \sum_{r\in\mathcal{R}_b} p_r \left| u_r h_r(\widehat{w}_b, \widehat{e}_b, u_r) - u_r h_r(\bar{w}_b^*, \bar{e}_b^*, u_r) \right| \\
&\le C|\widehat{\ell}_b - \ell_b^*| + \sum_{r\in\mathcal{R}_b} p_r \left(L_w|\widehat{w}_b - \bar{w}_b^*| + L_e|\widehat{e}_b - \bar{e}_b^*| \right) \\
&\le C\epsilon_\ell + \sum_{r\in\mathcal{R}_b} p_r(L_w\epsilon_w + L_e\epsilon_e) = B_b^{\text{pred}}.
\end{aligned}$$

The second inequality applies the revenue-sensitivity bounds (L_w, L_e) in Eq. (58) to each order, and the third substitutes the prediction-error bounds $(\epsilon_\ell, \epsilon_w, \epsilon_e)$ in Eq. (57). Thus, $C\epsilon_\ell$ bounds the discrepancy in travel cost, while the sum bounds the discrepancy in expected order revenue. This bound holds uniformly for every $\boldsymbol{u} \in \mathcal{U}_b$.

Recall that η_b and η_b^0 are the optimal values of f_b and f_b^0, respectively, over the same pricing domain $\mathcal{U}_b$. The preceding uniform bound implies, for every $\boldsymbol{u} \in \mathcal{U}_b$,

$$f_b(\boldsymbol{u}) \le f_b^0(\boldsymbol{u}) + B_b^{\text{pred}} \le \eta_b^0 + B_b^{\text{pred}}.$$

Thus, even the largest value of f_b cannot exceed $\eta_b^0 + B_b^{\text{pred}}$, giving $\eta_b \le \eta_b^0 + B_b^{\text{pred}}$. Reversing the roles of f_b and f_b^0 similarly gives $\eta_b^0 \le \eta_b + B_b^{\text{pred}}$. Combining these two inequalities yields

$$-B_b^{\text{pred}} \le \eta_b - \eta_b^0 \le B_b^{\text{pred}}, \quad \text{or equivalently} \quad |\eta_b - \eta_b^0| \le B_b^{\text{pred}},$$

which proves Eq. (S2). Combining it with $A_b = |v_b - \eta_b^0|$ and the triangle inequality proves Eq. (59).

Step 2: Bound the objective error for a feasible matching. For any feasible matching $\boldsymbol{\alpha} \in \mathcal{A}$, summing the pattern-level bounds over its selected patterns gives

$$|V(\boldsymbol{\alpha}) - \widehat{V}(\boldsymbol{\alpha})| \le \sum_{b\in\Omega} |v_b - \eta_b|\alpha_b \le \sum_{b\in\Omega} (A_b + B_b^{\text{pred}})\alpha_b = E(\boldsymbol{\alpha}), \qquad \boldsymbol{\alpha} \in \mathcal{A}.$$

Step 3: Bound the loss from matching selection. Apply Step 2 to the optimal matching $\boldsymbol{\alpha}^*$ and the returned matching $\widehat{\boldsymbol{\alpha}}$. Adding and subtracting their A-CAM profit values and using the optimization-gap bound in Eq. (60) gives

$$\begin{aligned} Z^* - V(\widehat{\boldsymbol{\alpha}}) &= \underbrace{V(\boldsymbol{\alpha}^*) - \widehat{V}(\boldsymbol{\alpha}^*)}_{\le E(\boldsymbol{\alpha}^*)} + \underbrace{\widehat{V}(\boldsymbol{\alpha}^*) - \widehat{V}(\widehat{\boldsymbol{\alpha}})}_{\le \Gamma} \\ &\quad + \underbrace{\widehat{V}(\widehat{\boldsymbol{\alpha}}) - V(\widehat{\boldsymbol{\alpha}})}_{\le E(\widehat{\boldsymbol{\alpha}})} \\ &\le E(\boldsymbol{\alpha}^*) + \Gamma + E(\widehat{\boldsymbol{\alpha}}). \end{aligned}$$

The first and third terms are signed errors in evaluating the O-CAM-optimal and A-CAM-selected matchings, respectively; they are not separate nonnegative losses. The middle term is bounded by the A-CAM optimality gap because

$$\widehat{V}(\boldsymbol{\alpha}^*) - \widehat{V}(\widehat{\boldsymbol{\alpha}}) \le \max_{\boldsymbol{\alpha} \in \mathcal{A}} \widehat{V}(\boldsymbol{\alpha}) - \widehat{V}(\widehat{\boldsymbol{\alpha}}) \le \Gamma.$$

Thus, $E(\boldsymbol{\alpha}^*)$ and $E(\widehat{\boldsymbol{\alpha}})$ bound the two valuation errors, while Γ allows for incomplete optimization of A-CAM. The second inequality in Eq. (61) follows by bounding each error sum by its maximum over $\mathcal{A}$. Nonnegativity follows from the optimality of $\boldsymbol{\alpha}^*$ for the exact objective. □

The practical evaluation of the bound is summarized in Appendix B.5.

S1.1.1. Size-dependent bound and scope. To examine how the optimality-loss bound changes as instance size increases, we express its prediction-error component in terms of the numbers of orders and vehicles, under the assumptions of Theorem 4. Let n and K denote these numbers, respectively, and let N be the maximum number of orders in a nonempty pooling pattern. Each selected pattern uses one vehicle, and neither a vehicle nor an order can appear in more than one selected pattern. Since each nonempty pattern contains at least one order, at most $M_{\text{pat}} = \min\{n, K\}$ such patterns can be selected. They can serve at most $J = \min\{n, NK\}$ distinct orders.

Assume that actions serving no orders are excluded or evaluated identically in both models. For identically evaluated actions, use $A_b = B_b^{\text{pred}} = 0$ in the error sums instead of the general bound in Eq. (S2). Let $0 \le p_r \le p_{\max}$ and $C \ge 0$. The largest total structural error over feasible matchings is

$$A_{\max} := \max_{\boldsymbol{\alpha} \in \mathcal{A}} \sum_{b \in \Omega} A_b \alpha_b.$$

Then the argument of Theorem 4 yields

$$Z^* - V(\widehat{\boldsymbol{\alpha}}) \le \Gamma + 2A_{\max} + 2CM_{\text{pat}}\epsilon_\ell + 2p_{\max}J(L_w\epsilon_w + L_e\epsilon_e). \tag{S3}$$

Derivation. The tour-length error contributes at most $C\epsilon_\ell$ per nonempty pattern. The errors in average waiting-for-pickup time and average detour ratio contribute at most $p_{\max}(L_w\epsilon_w + L_e\epsilon_e)$ per assigned order. Summing over at most M_{pat} patterns and J orders gives

$$E(\boldsymbol{\alpha}) = \sum_{b \in \Omega} (A_b + B_b^{\text{pred}})\alpha_b \le A_{\max} + CM_{\text{pat}}\epsilon_\ell + p_{\max}J(L_w\epsilon_w + L_e\epsilon_e), \qquad \boldsymbol{\alpha} \in \mathcal{A}.$$

Applying this estimate to both $E(\boldsymbol{\alpha}^*)$ and $E(\widehat{\boldsymbol{\alpha}})$ in Eq. (61) gives Eq. (S3), including the factor of two. □

For fixed C, $p_{\max}$, prediction-error bounds $(\epsilon_\ell, \epsilon_w, \epsilon_e)$, and sensitivity bounds (L_w, L_e), the prediction-error term scales linearly with M_{pat} and J. This statement concerns the bound's prediction-error term, not the actual loss or the other terms $A_{\max}$ and Γ. Even with exact predictions, structural error and an A-CAM optimization gap may remain.

Allowing for incomplete post-optimization. We extend the preceding bounds to the profit of the solution actually returned when post-optimization stops before reaching the fixed-matching optimum, for example because of a time limit. The selected matching $\widehat{\boldsymbol{\alpha}}$ remains fixed. Let Z^{post} denote the returned feasible profit, evaluated using the O-CAM objective, and recall that $V(\widehat{\boldsymbol{\alpha}})$ is the optimal profit attainable with this matching. The total loss can then be written as

$$\underbrace{Z^* - Z^{\text{post}}}_{\text{total loss}} = \underbrace{Z^* - V(\widehat{\boldsymbol{\alpha}})}_{\text{matching-selection loss}} + \underbrace{V(\widehat{\boldsymbol{\alpha}}) - Z^{\text{post}}}_{\text{post-optimization loss}}.$$

The first term is bounded by Eqs. (61) and (S3). If a valid upper bound Δ_{post} on the second term is available, adding it to either bound gives an upper bound on the total loss $Z^* - Z^{\text{post}}$. Such a bound can be obtained from a certified upper bound on the fixed-matching

routing-and-pricing optimum minus the feasible profit Z^{post}; it cannot be inferred from the feasible profit alone. When post-optimization is solved to optimality, $\Delta_{\text{post}} = 0$ and the original bounds apply without adjustment. □

Small-instance evaluation and scope. On sufficiently small instances, enumerate all feasible patterns and compute their profits using the procedure in Appendix B.5. Solving the exact and approximate pattern-selection problems gives α^* and an A-CAM-optimal $\widehat{\alpha}$, with $\Gamma = 0$. The resulting loss can then be compared directly with Eq. (61). Maximizing $\sum_b A_b \alpha_b$ over $\mathcal{A}$ also gives $A_{\max}$ for Eq. (S3). Appendix C.2 reports the corresponding numerical comparison of worst-case loss and the theoretical bound under prescribed prediction-error limits; see Fig. 9. This is an ex post assessment of the theoretical bound and its tightness, rather than evidence that the bound can be evaluated cheaply on large instances.

Comparison with the global O-CAM optimum requires a complete, common feasible pattern set and prediction-error bounds $(\epsilon_\ell, \epsilon_w, \epsilon_e)$ valid for every pattern. With a restricted pattern set and a gap Γ measured over that set, the bound applies only to its restricted O-CAM optimum. For larger instances, evaluating the bound requires valid structural-error and optimization-gap bounds in place of exhaustive calculation.

S1.2. Proof of the strong NP-hardness of the order cancellation-aware on-demand ride-pooling problem

Proof. The strong NP-hardness of the studied order cancellation-aware on-demand ride-pooling problem can be proven via a polynomial-time reduction from a known strongly NP-hard problem.

First, as stated in Section 1, the studied problem generalizes the classical multi-vehicle static Dial-a-Ride Problem (DARP), in which passenger requests are served by a fleet of vehicles subject to time window, capacity, and pickup and delivery constraints.

Compared to the standard multi-vehicle static DARP, the O-CAM model introduces additional complexities, including the pricing decisions u_r, cancellation-related decision variables h_r and nonlinear constraints in Eq. (1), as well as the anticipatory repositioning decisions associated with predicted future orders.

The O-CAM model can be transformed as the standard multi-vehicle static DARP via the following polynomial time transformation procedures: i) Setting $\mathcal{R}_1 = \mathcal{R}_2 = \mathcal{R}^F = \emptyset$; ii) Fixing the pricing decisions $u_r = 1, \forall r \in \mathcal{R}_3$; iii) Removing the constraints in Eq. (1), and iv) Setting the acceptance decisions $h_r = 1, \forall r \in \mathcal{R}_3$.

According to Savelsbergh (1985), Cordeau (2006), even the single-vehicle static DARP is a generalization of the TSPTW, which is strongly NP-hard. As a result, the multi-vehicle static DARP is also strongly NP-hard. Because the O-CAM model generalizes the multi-vehicle static DARP, it is also strongly NP-hard.

□

S1.3. Proof of the NP-hardness of the A-CAM model

Proof. We prove the NP-hardness of the A-CAM model via reformation-based reduction.

As shown in Section 4.3.2, A-CAM can be equivalently reformulated as a *Maximum Weighted Set Packing (MWSP)* problem defined over the set of all feasible pooling patterns Ω. Each element $b \in \Omega$ represents a feasible combination of orders assigned to a vehicle, along with their associated optimized pricing. The reformulated model is given by Equations (47) to (50).

Each pooling pattern involves selecting a subset of riders whose group ride is feasible. For each driver $k \in \mathcal{K}$, the total number of such feasible groupings is bounded above by $O(|\mathcal{R}|^N)$, where N is the maximum number of orders that can be assigned to a vehicle. Thus, the total number of feasible pooling patterns is $O(|\mathcal{K}| \cdot |\mathcal{R}|^N)$, which is polynomial in $|\mathcal{R}|$ when N is fixed.

Despite $|\Omega|$ being polynomial in $|\mathcal{R}|$ for fixed $N(\geqslant 3)$, the reformulated problem is still a MWSP problem—a well-known NP-hard combinatorial optimization problem (Garey et al., 2002). MWSP is NP-hard even when the size of each set (corresponds to a pooling pattern in Ω) is bounded by a small constant (e.g., 3), as it generalizes the NP-complete 3-Dimensional Matching (3DM) problem (Karp, 1972).

Specifically, we can reduce 3DM to a special case of the A-CAM model based on the reformulated model (i.e., Equations (47) to (50)): i) Each triplet in 3DM corresponds to a feasible pooling pattern b in the reformulated model, involving two riders matched to one driver; ii) The constraints in Eqs. (48) and (49) ensure that each rider and driver appears in at most one selected pattern; iii) The goal is to maximize the total weight (or number) of selected disjoint triplets/patterns—the same as in 3DM. Hence, despite $|\Omega|$ being polynomially increasing in $|\mathcal{R}|$ for fixed $N(\geqslant 3)$, the set packing structure and disjointness constraints make the problem NP-hard. □

S2. Worst-Case Loss Evaluation: Formulation and Derivation

We numerically validate the bound in Theorem 4 by comparing it with worst-case optimality losses under prescribed prediction-error limits. Small instances allow complete pattern enumeration and exact optimization, so both the loss and the bound can be evaluated. This provides an ex post assessment of the bound's tightness across error levels and instance sizes.

For instance i, let Ω_i be the complete feasible pooling-pattern set and $\mathcal{A}_i$ the corresponding set of feasible matchings. For each $b \in \Omega_i$, compute v_b by solving the single-vehicle RPP under Policy 2, as described in Appendix B.5. Define

$$V(\boldsymbol{\alpha}) = \sum_{b \in \Omega_i} v_b \alpha_b, \qquad \boldsymbol{\alpha}_i^* \in \arg\max_{\boldsymbol{\alpha} \in \mathcal{A}_i} V(\boldsymbol{\alpha}), \qquad Z_i^* = V(\boldsymbol{\alpha}_i^*).$$

For each admissible predictor realization, A-CAM determines a matching; its profit is then evaluated by V, with routing and pricing optimized exactly for that fixed matching. We seek the matching with the lowest such profit among those that can be optimal for A-CAM. If several matchings are A-CAM-optimal, the worst-case comparison includes the one with the lowest V.

Selection of error bounds and sensitivity constants. The radii $(\epsilon_\ell, \epsilon_w, \epsilon_e)$ specify the prediction-error limits; (L_w, L_e) bound the sensitivity of normalized expected revenue to pickup time and detour ratio. We determine these two sets of inputs separately.

We use the 95% split-conformal calibration radii from Appendix C.1 as base error scales. For predictor $j \in \{\ell, w, e\}$ and m exchangeable calibration residuals $r_{sj} = |\widehat{x}_{sj} - x_{sj}|$, the radius is the $\lceil 0.95(m+1) \rceil$-th smallest residual (infinite if this index exceeds m). Residuals must match the targets in Eq. (57); tour-length units must also match the cost coefficient C. These marginal radii set the perturbation ranges used below, rather than guarantee simultaneous coverage of all patterns. The theorem applies conditionally within the prescribed ranges.

To determine the sensitivity constants, write $g_r(w, e, u) = u h_r(w, e, u)$. Uniform bounds on $|\partial g_r / \partial w|$ and $|\partial g_r / \partial e|$ over the relevant domain yield Eq. (58). For the fitted function in Eq. (62), write $h_r = \mathrm{clip}(q_r, 0, 1)$, where q_r is the expression before truncation. Its coefficients of w and e are

$$a_w(u) = -0.015 + 0.011(1-u), \qquad a_e(u, W_r) = -0.408 + 0.050(1-u)W_r.$$

Since clipping is 1-Lipschitz, even at its nondifferentiable breakpoints,

$$|g_r(w, e, u) - g_r(w', e', u)| \le u|a_w(u)|\,|w - w'| + u|a_e(u, W_r)|\,|e - e'|.$$

Under the experimental ranges $u \in [0, 1]$ and $W_r \in [0, 1]$, these coefficients satisfy $u|a_w(u)| \le 0.015$ and $u|a_e(u, W_r)| \le 0.408$. Thus, the settings

$$L_w = 0.015, \qquad L_e = 0.408$$

are valid uniform bounds for normalized expected revenue, including the pricing factor u. For monetary revenue, these bounds are multiplied by p_r, as in B_b^{pred}. They must be reassessed if the acceptance function, its fitted coefficients, or the admissible parameter ranges change.

Prediction-error levels. We use the scalar multiplier ρ to control the magnitude of prediction-error perturbations, while $\boldsymbol{\alpha}$ remains the pattern-selection vector. We consider $\rho \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}$. Here, $\rho = 0$ gives no prediction perturbation, $\rho = 1$ permits errors up to the prescribed base bounds, and intermediate values scale these limits proportionally. Specifically, $|\widehat{\ell}_b - \ell_b^*| \le \rho\epsilon_\ell$, $|\widehat{w}_b - \bar{w}_b^*| \le \rho\epsilon_w$, and $|\widehat{e}_b - \bar{e}_b^*| \le \rho\epsilon_e$. The prediction-error radii used in the numerical analysis are $\rho \times (6.09, 7.4444, 0.0949)$ in the computational units for tour length, pickup time, and detour ratio, respectively. Appendix C.1 reports the corresponding 95% calibration scales as $(2.4988, 7.4444, 0.0949)$; the experiment configuration pairs the reported tour-length scale 2.4988 with 6.09 code units. The numerical calculations use $C = 0.17$ per tour-code unit. The multiplier ρ scales these base radii.

Worst-case optimality-loss model. For each fixed pattern b, let $(\ell_b^*, \bar{w}_b^*, \bar{e}_b^*)$ be the exact tour length, average waiting-for-pickup time, and average detour ratio obtained from its optimal single-vehicle RPP solution. These are fixed-pattern reference values, not the attributes of a single globally optimal matching. Let $\boldsymbol{\xi}_b = (\xi_{\ell b}, \xi_{wb}, \xi_{eb})$ denote signed perturbations satisfying

$$\begin{aligned}\mathcal{U}_b(\rho) := \{\boldsymbol{\xi}_b : {} & |\xi_{\ell b}| \le \epsilon_\ell,\ |\xi_{wb}| \le \epsilon_w,\ |\xi_{eb}| \le \epsilon_e, \\ & \ell_b^* + \rho\xi_{\ell b} \ge 0,\ \bar{w}_b^* + \rho\xi_{wb} \ge 0,\ \bar{e}_b^* + \rho\xi_{eb} \ge 0\}, \\ \mathcal{U}_i(\rho) := {} & \prod_{b \in \Omega_i} \mathcal{U}_b(\rho).\end{aligned}$$

The nonnegativity restriction gives the admissible ranges $\widehat{\ell}_b \in [\max\{0, \ell_b^* - \rho\epsilon_\ell\}, \ell_b^* + \rho\epsilon_\ell]$, $\widehat{w}_b \in [\max\{0, \bar{w}_b^* - \rho\epsilon_w\}, \bar{w}_b^* + \rho\epsilon_w]$, and $\widehat{e}_b \in [\max\{0, \bar{e}_b^* - \rho\epsilon_e\}, \bar{e}_b^* + \rho\epsilon_e]$, consistent with the implementation. Empty patterns are evaluated identically in both models, with zero profit, zero perturbation, and $A_b = B_b^{\text{pred}} = 0$. At error level ρ, A-CAM uses the perturbed outputs $\widehat{\ell}_b(\boldsymbol{\xi}_b; \rho) = \ell_b^* + \rho\xi_{\ell b}$, $\widehat{w}_b(\boldsymbol{\xi}_b; \rho) = \bar{w}_b^* + \rho\xi_{wb}$, and $\widehat{e}_b(\boldsymbol{\xi}_b; \rho) = \bar{e}_b^* + \rho\xi_{eb}$. Thus, $\boldsymbol{\xi}_b$ denotes perturbations, while $(\widehat{\ell}_b, \widehat{w}_b, \widehat{e}_b)$ are the perturbed values. The actual prediction errors are $(\rho\xi_{\ell b}, \rho\xi_{wb}, \rho\xi_{eb})$, which satisfy the limits stated above. At $\rho = 0$, all three predictor outputs equal their exact reference values.

For a given perturbation, the A-CAM value is obtained by substituting these outputs into the pricing problem:

$$\eta_b(\boldsymbol{\xi}_b; \rho) = \max_{\boldsymbol{u} \in [0,1]^{|\mathcal{R}_b|}} \Big\{ \sum_{r \in \mathcal{R}_b} p_r u_r h_r(\bar{w}_b^* + \rho\xi_{wb}, \bar{e}_b^* + \rho\xi_{eb}, u_r) - C(\ell_b^* + \rho\xi_{\ell b}) \Big\}.$$

In particular, $\eta_b(\boldsymbol{\xi}_b; 0) = \eta_b^0$: zero prediction error recovers the exact-average A-CAM value, which need not equal v_b. Thus, prices are optimized for each predictor realization. With these pattern values, A-CAM selects a matching that maximizes approximate profit. Intuitively, the perturbations change how attractive each matching appears to A-CAM, while its exact profit remains unchanged. We ask how far these changes can steer A-CAM toward a matching that looks optimal under the perturbed predictions but performs poorly after exact routing and pricing optimization. The following bilevel model formalizes this question:

$$\begin{aligned} L_i^{\text{wc}}(\rho) = \max_{\boldsymbol{\xi}, \widehat{\boldsymbol{\alpha}}} \quad & Z_i^* - V(\widehat{\boldsymbol{\alpha}}), \quad \boldsymbol{\xi} \in \mathcal{U}_i(\rho), \\ \text{s.t.} \quad & \widehat{\boldsymbol{\alpha}} \in \arg\max_{\boldsymbol{\alpha} \in \mathcal{A}_i} \sum_{b \in \Omega_i} \eta_b(\boldsymbol{\xi}_b; \rho)\alpha_b. \end{aligned}$$

Here, $\boldsymbol{\alpha} = (\alpha_b)_{b \in \Omega_i} \in \mathcal{A}_i$ is the pattern-selection vector, using the notation of Section 4.3.2: $\alpha_b = 1$ if pattern b is selected and $\alpha_b = 0$ otherwise. The set $\mathcal{A}_i$ enforces feasibility of the selected patterns. The vector $\widehat{\boldsymbol{\alpha}}$ is a maximizing solution of this inner A-CAM problem for the given perturbed predictions. The outer problem chooses perturbations $\boldsymbol{\xi} \in \mathcal{U}_i(\rho)$, and hence the perturbed predictor outputs, that make the selected matching's exact profit $V(\widehat{\boldsymbol{\alpha}})$ as small as possible. Since Z_i^* is fixed, minimizing that profit is equivalent to maximizing the optimality loss. If the inner problem has multiple optimal matchings, the outer problem selects the one with the lowest exact profit; this is the pessimistic tie-breaking convention used here. The exact pattern profits v_b and Z_i^* remain fixed throughout the perturbation analysis.

Solving the worst-case model. Rather than searching directly over all combinations of continuous perturbations, we examine candidate matchings. For each candidate, we ask whether any allowed perturbation could make A-CAM select it. This turns the worst-case search into a sequence of matching-selection tests on small instances. For each pattern, first compute the attainable profit bounds

$$\underline{\eta}_b(\rho) = \min_{\boldsymbol{\xi}_b \in \mathcal{U}_b(\rho)} \eta_b(\boldsymbol{\xi}_b; \rho), \qquad \overline{\eta}_b(\rho) = \max_{\boldsymbol{\xi}_b \in \mathcal{U}_b(\rho)} \eta_b(\boldsymbol{\xi}_b; \rho).$$

For the fitted acceptance function and $W_r \in [0, 1]$, optimized profit is nonincreasing in each predictor input. Thus, the upper profit is attained at the three lower predictor endpoints and the lower profit at the three upper endpoints. This is the endpoint calculation used in the code. These bounds describe the lowest and highest profit that A-CAM can assign to pattern b as its predictions vary within the allowed range, with prices reoptimized each time. For a candidate matching $\boldsymbol{a} \in \mathcal{A}_i$, form

$$\eta_b^{\boldsymbol{a}}(\rho) = \begin{cases} \overline{\eta}_b(\rho), & a_b = 1, \\ \underline{\eta}_b(\rho), & a_b = 0. \end{cases}$$

This vector favors the candidate by assigning the highest attainable profits to its selected patterns and the lowest to all other patterns. Under the independent patternwise box set $\mathcal{U}_i(\rho)$, the candidate can be A-CAM-optimal for some admissible predictions if and only if

$$\sum_{b \in \Omega_i} \eta_b^{\boldsymbol{a}}(\rho) a_b = \max_{\boldsymbol{\alpha} \in \mathcal{A}_i} \sum_{b \in \Omega_i} \eta_b^{\boldsymbol{a}}(\rho)\alpha_b.$$

We call a matching satisfying this test *inducible*, and denote the set of such matchings by $\mathcal{A}_i^{\text{ind}}(\rho)$. Intuitively, the test gives the candidate its best possible chance of being selected. If it is still not optimal for A-CAM, no other allowed perturbation can make it optimal. If it passes, it can indeed be selected under an admissible perturbation, including a tie resolved in its favor.

The following proposition justifies using these tests to solve the worst-case model: among matchings that can be selected, the one with the lowest exact profit produces the largest loss.

Proposition 1 (Equivalent characterization of worst-case optimality loss) *Under the patternwise box uncertainty set and pessimistic tie-breaking convention above,*

$$L_i^{\mathrm{wc}}(\rho) = Z_i^* - \min_{\boldsymbol{a} \in \mathcal{A}_i^{\mathrm{ind}}(\rho)} V(\boldsymbol{a}).$$

Consequently, examining feasible matchings in nondecreasing order of V identifies a worst-case matching at the first successful inducibility test.

Proof. For any competing matching $\boldsymbol{\alpha}$, the candidate's approximate-profit advantage is $\sum_b \eta_b(a_b - \alpha_b)$. Raising coefficients with $a_b = 1$ and lowering those with $a_b = 0$ cannot reduce this advantage. Hence, if the candidate is optimal for any admissible coefficient vector, it is optimal for the candidate-favoring vector. Conversely, the patternwise product set allows the selected extrema to be attained simultaneously, so optimality under this vector establishes inducibility. The matchings satisfying the inner optimality constraint for at least one admissible perturbation are therefore exactly $\mathcal{A}_i^{\mathrm{ind}}(\rho)$. Maximizing $Z_i^* - V$ over this set gives the stated expression and the ordered-search rule. □

Accordingly, we enumerate $\mathcal{A}_i = \{\boldsymbol{\alpha}^{(1)}, \ldots, \boldsymbol{\alpha}^{(N_i)}\}$ with

$$V(\boldsymbol{\alpha}^{(1)}) \le \cdots \le V(\boldsymbol{\alpha}^{(N_i)}),$$

and apply the test sequentially. The first inducible matching, denoted by $\widehat{\boldsymbol{\alpha}}_i^{\mathrm{wc}}$, solves the worst-case model when all preceding candidates have been ruled out without search truncation and the required optimization problems are solved to optimality. All 180 runs were solved to optimality.

Comparison with the theoretical bound. We evaluate ten fixed instances from each of A_{4_2}, A_{6_3}, and A_{8_4}, yielding 30 instances in total. These instances are distinct from those used in the main numerical experiments. For each instance and error level, the identified worst-case optimality loss is

$$L_i^{\mathrm{wc}}(\rho) = Z_i^* - V(\widehat{\boldsymbol{\alpha}}_i^{\mathrm{wc}}),$$

and the corresponding theoretical optimality-loss bound is

$$B_i^{\mathrm{sel}}(\rho) = \Gamma_i + E(\boldsymbol{\alpha}_i^*; \rho) + E(\widehat{\boldsymbol{\alpha}}_i^{\mathrm{wc}}; \rho),$$

where $E(\boldsymbol{\alpha}; \rho)$ bounds the absolute difference between the O-CAM and A-CAM profit evaluations of the same matching $\boldsymbol{\alpha}$ when the prediction-error limits are $(\rho\epsilon_\ell, \rho\epsilon_w, \rho\epsilon_e)$. It sums the structural and prediction-error bounds over the selected patterns:

$$\begin{aligned} E(\boldsymbol{\alpha}; \rho) &= \sum_{b \in \Omega_i} (A_b + \rho B_b^{\mathrm{pred}})\alpha_b \\ &= \sum_{b \in \Omega_i} \left[A_b + \rho \left(C\epsilon_\ell + \sum_{r \in \mathcal{R}_b} p_r (L_w \epsilon_w + L_e \epsilon_e) \right) \right] \alpha_b . \end{aligned}$$

Here, $A_b = |v_b - \eta_b^0|$ is the structural profit discrepancy; only the prediction-error term B_b^{pred} is scaled by ρ. The two E terms correspond to the globally optimal O-CAM matching and the identified worst-case A-CAM matching. The term Γ_i bounds the A-CAM optimization gap and is zero for an exact solve.

To compare instances with different profit levels, we report relative optimality loss and its upper bound, both normalized by the positive O-CAM optimum Z_i^*:

$$\text{Relative loss}_i(\rho) = 100 \frac{L_i^{\mathrm{wc}}(\rho)}{Z_i^*}, \qquad \text{Relative bound}_i(\rho) = 100 \frac{B_i^{\mathrm{sel}}(\rho)}{Z_i^*}.$$

Both quantities are percentages; they measure loss in optimized profit, not relative prediction error.

S3. Sensitivity Analysis of Vehicle-Average and Order-Specific Predictors

The pickup-time and detour-ratio predictors used in the main framework adopt vehicle-pool average targets to preserve the route-aggregated, pre-routing structure of A-CAM. Because this aggregation may omit within-pool heterogeneity, we additionally test direct order-specific linear predictors using only information available before route sequencing. The complete predictor construction, instance-level results, and computational details are provided in the online supplementary material available at https://github.com/data-sharing-user/order-specific-predictor-sensitivity-supplement.

Table S1 Order-level predictive performance of vehicle-average and order-specific linear predictors

Attribute	Predictor	MAE	RMSE
Pickup time	Vehicle-average	9.007	12.482
Pickup time	Order-specific	8.103	11.081
Detour ratio	Vehicle-average	0.083	0.132
Detour ratio	Order-specific	0.059	0.096

Relative to assigning the vehicle-average prediction to each order, the order-specific predictor reduces pickup-time MAE and RMSE by 10.04% and 11.22%, respectively, and reduces detour-ratio MAE and RMSE by 29.08% and 27.56%, respectively. These results confirm that pre-routing order-level features capture part of the within-pool heterogeneity.

Table S2 Summary of downstream sensitivity to order-specific predictors

Framework	Mean ΔObj (%)	Mean ΔTime (%)	Higher / Equal / Lower	Additional model-size effect
A-CAM+PO	0.88	46.30*	23 / 12 / 15	+30.03% variables; +25.05% constraints
PAD^2	2.49	4.70	30 / 11 / 9	Procedural predictor evaluation

Notes: Each framework is evaluated on 50 paired cases from five instance groups. *For A-CAM+PO, the mean paired runtime increase is calculated from the 20 uncensored A_{4_2} and A_{6_3} cases; at A_{8_4} and larger groups, at least one formulation reaches the 60-second limit. "Higher / Equal / Lower" reports the number of cases in which the order-specific specification yields a higher, equal, or lower postprocessed objective than the vehicle-average specification.

For A-CAM+PO, the order-specific specification improves the postprocessed objective by only 0.88% on average while substantially increasing model size and computation time. The computational effect becomes more pronounced as instance size increases, and the objective improvement is not uniformly positive across cases. In contrast, PAD^2 can incorporate the same order-specific predictors procedurally, yielding a larger average objective improvement of 2.49% with a relatively small average runtime increase of 4.70%.

The order-specific predictors require richer tailored features; embedding them in A-CAM adds auxiliary variables and constraints and increases solution time. Given the modest average objective improvement and the additional computational cost, we retain vehicle-average predictors in the main A-CAM formulation to balance predictive accuracy and computational efficiency. Order-specific predictors remain a useful extension of PAD^2, where they can be evaluated procedurally at comparatively modest computational cost. Detailed results for all five instance groups are reported in the online supplementary material.

S4. Algorithmic Procedures and Benchmark Implementation

This section gives the column-generation procedure and the ALNS benchmark implementation. Graph reduction and initialization are provided in Appendix H.

S4.1. Column Generation Procedure

Algorithm S1 alternates between solving the LP relaxation of the RMP and solving the vehicle-specific pricing subproblems SP(k), $k \in \mathcal{K}$, to generate additional pooling patterns with positive reduced cost. Graph reduction is applied once to obtain the candidate order sets $\{\mathcal{A}_k : k \in \mathcal{K}\}$, and Algorithm 2 supplies the initial columns.

At each iteration, the LP relaxation of the RMP yields the dual values for the order and vehicle constraints. For each vehicle, the pricing step examines candidate order groups, evaluates their predicted service measures, and optimizes order prices to determine each pattern's profit. Patterns with positive reduced cost are added to the RMP. The iterations stop when no improving column is found or the time limit is reached. Finally, the RMP is solved with binary selection variables to obtain a feasible matching from the accumulated columns.

Measured initial-column counts. Table S3 reports the average number of RMP columns generated during the initial-column construction stage of L-PAD^2 under the computational time limit. Each row reports the arithmetic mean over the five cases in the corresponding instance group. Counts include columns generated before termination when the time limit interrupts initialization. Detailed instance-level data are available in the initial RMP column-count data repository. These are counts of available columns, rather than counts of selected patterns or retained vehicle–order pairs. Several alternative columns may correspond to the same vehicle; the RMP subsequently selects at most one pattern per vehicle.

Table S3 Average number of initial RMP columns for L-PAD^2

Instance group	Initial columns
A_4_2	14.4
A_6_3	114.2
A_8_4	490.2
A_10_5	1467.6
A_15_8	2773.0
B_20_10	4522.2
B_25_10	5186.6
B_30_10	8168.8
B_40_15	5311.0
B_50_20	8846.0
C_60_20	11574.8
C_70_20	12337.0
C_80_25	20273.2
C_90_25	12855.4
C_100_25	5945.4

Note: Each row reports the arithmetic mean over the five cases in the corresponding instance group. Counts include columns generated before termination when the time limit interrupts initialization. Detailed instance-level data are available in the initial RMP column-count data repository.

Algorithm S1 Column-generation procedure

Require: Order set $\mathcal{R}_3$, dispatchable vehicle set $\mathcal{K}$, maximum pool size N, and time limit $T_{\max}$

Ensure: Selected pooling patterns and corresponding decisions

Apply Algorithm 1 to obtain $\{\mathcal{A}_k : k \in \mathcal{K}\}$
Apply Algorithm 2 to generate $\Omega'^{(0)}$
Set $t \leftarrow 0$
repeat
  Solve the LP relaxation of RMP($\Omega'^{(t)}$)
  Obtain dual values $\lambda_r^{(t)}$, $r \in \mathcal{R}_3$, and $\delta_k^{(t)}$, $k \in \mathcal{K}$, associated with constraints (48) and (49)
  Set $\mathcal{B}_+^{(t)} \leftarrow \emptyset$
  for $k \in \mathcal{K}$ **do**
    Enumerate $S \subseteq \mathcal{A}_k$ satisfying $1 \le |S| \le N$
    for each candidate order group S **do**
      Let $\boldsymbol{\beta}^S$ be the incidence vector of S
      Predict $\widetilde{w} \leftarrow f^T(\boldsymbol{\beta}^S)$ and $\widetilde{e} \leftarrow f^E(\boldsymbol{\beta}^S)$
      for $r \in S$ **do**
        Determine $u_r^* \leftarrow \arg\max_{0 \le u_r \le 1} u_r F(u_r, \widetilde{w}, \widetilde{e} \mid W_r, E_r)$
        Set $h_r \leftarrow F(u_r^*, \widetilde{w}, \widetilde{e} \mid W_r, E_r)$
      end for
      Predict the P&D tour length $l \leftarrow f^D(\boldsymbol{\beta}^S)$
      Define column b by k, S, and $\{u_r^* : r \in S\}$
      Compute $\eta_b \leftarrow \sum_{r \in S} h_r p_r u_r^* - Cl$
      Compute $\bar{\eta}_b^{(t)} \leftarrow \eta_b - \sum_{r \in \mathcal{R}_3} \lambda_r^{(t)} \mu_{rb} - \sum_{k' \in \mathcal{K}} \delta_{k'}^{(t)} \pi_{k'b}$
      if $\bar{\eta}_b^{(t)} > 0$ and $b \notin \Omega'^{(t)}$ **then**
        $\mathcal{B}_+^{(t)} \leftarrow \mathcal{B}_+^{(t)} \cup \{b\}$
      end if

end for

end for

$\Omega'^{(t+1)} \leftarrow \Omega'^{(t)} \cup \mathcal{B}_+^{(t)}$

Set $t \leftarrow t+1$

until $\mathcal{B}_+^{(t-1)} = \emptyset$ or elapsed time reaches $T_{\max}$

Solve RMP($\Omega'^{(t)}$) with $\alpha_b \in \{0,1\}$ for all $b \in \Omega'^{(t)}$

return Selected pooling patterns and corresponding decisions

S4.2. Adaptive Large Neighborhood Search with Embedded Pricing

The adaptive large neighborhood search (ALNS) benchmark combines changes to order assignments and vehicle routes with an embedded pricing update. This allows the evaluation of a modified matching and route plan to account for its associated pricing decisions. The solution representation, destroy–repair operators, adaptive operator selection, acceptance criterion, and stopping rules are documented in the ALNS implementation supplement.

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